

HouseHunt

Premium House Rental Marketplace

Final Project Demonstration Report

Date	18 July 2026
Team ID	
Project Name	HouseHunt – Premium House Rental Marketplace
Team Leader	Shaik Mohammad Sameer
Team Members	Vamsi Talari, Chanukya Naidu Abothula, Karri Rajeswara Reddy, Revathi Devi Kota
Technology Stack	MongoDB · Express.js · React.js (Vite) · Node.js (MERN)

1. Introduction

1.1 Project Overview

HouseHunt is a full-stack MERN (MongoDB, Express.js, React.js, Node.js) web application that digitizes the house rental process by connecting renters, property owners and administrators through dedicated, role-based dashboards. It covers the complete rental journey — searching, listing, touring, negotiating, booking, paying and managing rental properties — on a single platform.

The application is built as a three-tier system: a React + Vite client for the presentation layer, an Express.js/Node.js REST API for business logic, and MongoDB (via Mongoose) for persistent data storage. It is deployed on Vercel, with the client served as a static site and the server running as a serverless API connected to a managed MongoDB cluster.

1.2 Purpose

HouseHunt aims to simplify the interaction between renters, property owners and administrators by replacing manual, paperwork-heavy rental workflows with a centralized digital platform. Renters can discover verified properties and communicate directly with owners; owners can manage listings, bookings and income efficiently; and administrators retain full oversight of verification, quality control and platform moderation. Traditional rental systems suffer from fake advertisements, manual booking procedures, delayed communication, poor record-keeping and a lack of centralized administration — HouseHunt addresses this gap with an integrated platform that securely connects all three stakeholder groups.

2. Ideation Phase

2.1 Problem Statement

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a renter searching for a home to rent	find a verified property that matches my budget and preferences and book a tour quickly	I can't tell which listings are genuine, and communication with owners is slow and inconsistent	the rental process is manual, listings aren't verified, and there's no centralized way to track my applications	frustrated and hesitant to commit to a booking
PS-2	a property owner listing my house for rent	advertise my property, manage tour requests and bookings, and collect rent efficiently	I have no centralized way to track requests, negotiate rent, or record commission and GST on payments	existing rental workflows rely on paperwork, phone calls and scattered spreadsheets	overwhelmed and unable to track my rental income accurately

2.2 Empathy Map Canvas

Persona: The Renter

Dimension	Insights
Think & Feel	Worried about scams and fake listings; wants a quick, trustworthy way to find a place that fits their budget and lifestyle.
See	Scattered listings across multiple sites/apps, inconsistent photos and pricing, unclear owner contact details.
Hear	Friends' rental horror stories about deposits and misleading ads; landlords who are slow to respond.
Say & Do	Compares multiple listings, messages owners repeatedly for details, delays committing until a tour is arranged.
Pain	Fake ads, manual booking, delayed communication, no centralized record of applications or payments.
Gain	Verified listings, transparent pricing and

	negotiation, a single dashboard to track every booking.
--	---

2.3 Brainstorming

Ideas generated during ideation, grouped and prioritized by importance and feasibility:

- Role-based dashboards for renter, owner and administrator (High importance, High feasibility)
- Verified-listing workflow with admin approval before publishing (High importance, High feasibility)
- Rent negotiation & counter-offer system (High importance, Medium feasibility)
- Real-time chat between renter and owner (Medium importance, Medium feasibility)
- Wallet with automatic GST & commission calculation (High importance, Medium feasibility)
- AI-assisted listing description validation & spam scoring (Medium importance, Medium feasibility)
- Google Maps integration for live location previews (Medium importance, High feasibility)
- CMS-driven homepage content, FAQ & announcements (Low importance, High feasibility)
- Native mobile app and payment gateway integration (High importance, Low feasibility — parked for future scope)

3. Requirement Analysis

3.1 Customer Journey Map

Renter journey — Browsing, touring, booking and reviewing a rental property:

Stage	Entice	Enter	Engage	Exit	Extend
What happens	Discovers HouseHunt via search or referral	Registers / signs in and lands on the discovery page	Searches, filters, views a listing, chats with the owner, negotiates rent and books a tour	Tour completes; renter confirms or cancels the booking	Leaves a review; receives personalized recommendations for future searches
Renter goal	"Help me find a place I can trust"	"Help me get started quickly"	"Help me compare and decide confidently"	"Help me know the booking went through"	"Help me remember and repeat a good experience"

3.2 Solution Requirement (Functional & Non-Functional)

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	User Registration	Registration through Form / Gmail / LinkedIn
FR-2	User Confirmation	Confirmation via Email / OTP
FR-3	Property Management	Add/edit/delete listing; publish or draft; image & amenity upload; admin approval before going live
FR-4	Booking & Negotiation	Submit/confirm/cancel tour requests; submit/counter-offer/accept negotiations
FR-5	Wallet & Payments	View wallet ledger; owner withdrawal requests; automatic GST & commission calculation
FR-6	Chat & Complaint Management	Real-time chat between renter and owner; register/track/resolve complaints

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
---------	----------------------------	-------------

NFR-1	Usability	Responsive UI (Tailwind CSS, Material UI, Bootstrap) with role-based dashboards across desktop and mobile.
NFR-2	Security	JWT-based authentication, bcryptjs password hashing, and role-guard middleware on every protected route.
NFR-3	Reliability	Mongoose schema validation and centralized Express error-handling middleware maintain data integrity.
NFR-4	Performance	Vite-based fast builds, an Axios API layer and indexed MongoDB queries keep interactions responsive.
NFR-5	Availability	Deployed on Vercel (static client + serverless API) backed by a managed MongoDB cluster.
NFR-6	Scalability	Modular three-tier architecture allows new modules (payments, mobile apps) to be added without reworking the core.

3.3 Data Flow Diagram

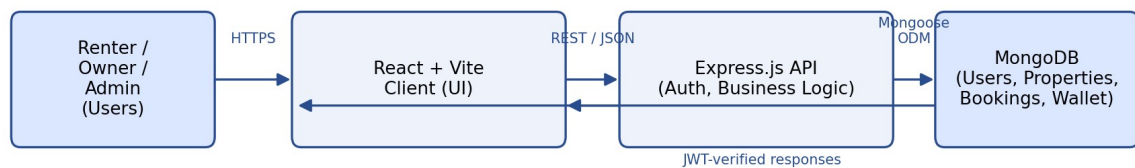


Figure: DFD Level-0 — Users interact with the React client, which calls the Express REST API; the API (after JWT verification) reads/writes MongoDB via Mongoose.

Representative user stories:

User Type	Functional Requirement	User Story	Priority
-----------	------------------------	------------	----------

Renter (Mobile/Web)	Registration	As a user, I can register by entering email, password, and confirming my password.	High
Renter (Web)	Property Search	As a renter, I can search and filter properties by location, price, furnishing and amenities.	High
Renter	Dashboard	As a renter, I can view visits requested, confirmed visits and pending answers on my dashboard.	Medium
Property Owner	Listing Management	As an owner, I can add, edit, publish or draft property listings with images and amenities.	High
Administrator	Property Moderation	As an admin, I can review and approve or reject submitted listings before they go live.	High
Administrator	Complaint Management	As an admin, I can view, prioritize and resolve tenant-reported complaints from a single console.	Medium

3.4 Technology Stack

Component	Technology
User Interface	React.js (Vite), Tailwind CSS, Material UI, Bootstrap
Application Logic	Node.js, Express.js
Authentication & Authorization	JWT, bcryptjs, role-guard middleware
Database	MongoDB (Mongoose ODM), MongoDB Atlas
External API	Google Maps API (live location previews & GPS-based search)
AI-assisted Moderation	Listing description validation & spam scoring service
Infrastructure	Vercel (static client + serverless API)

4. Project Design

4.1 Problem – Solution Fit

Renters and owners face a fragmented, low-trust rental process — fake listings, manual bookings, and no shared record of communication or payments. HouseHunt fits this problem by giving both sides a single, verified, role-based platform: owners get a structured way to list and manage properties, renters get verified listings with transparent negotiation, and administrators provide the trust layer through listing approval, complaint resolution and payment oversight — directly addressing the frequent annoyances (fraud, delay, poor record-keeping) that make the existing process painful.

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement	Fake/unverified listings, manual booking, delayed communication, poor record-keeping and no centralized administration.
2	Idea / Solution	An integrated MERN marketplace connecting renters, owners and admins via role-based dashboards covering search, listing, touring, negotiation and management.
3	Novelty / Uniqueness	Rent negotiation & counter-offer workflow, real-time chat, wallet with GST/commission tracking, AI-assisted listing moderation.
4	Social Impact / Satisfaction	Verified, admin-approved listings and a dedicated complaint centre reduce fraud and build renter/owner trust.
5	Business Model	Platform commission and GST on each completed transaction, tracked automatically in the wallet ledger.
6	Scalability	Modular three-tier MERN architecture with RBAC, deployed serverless on Vercel, ready for payment gateway, mobile apps and AI

		recommendations.
--	--	------------------

4.3 Solution Architecture

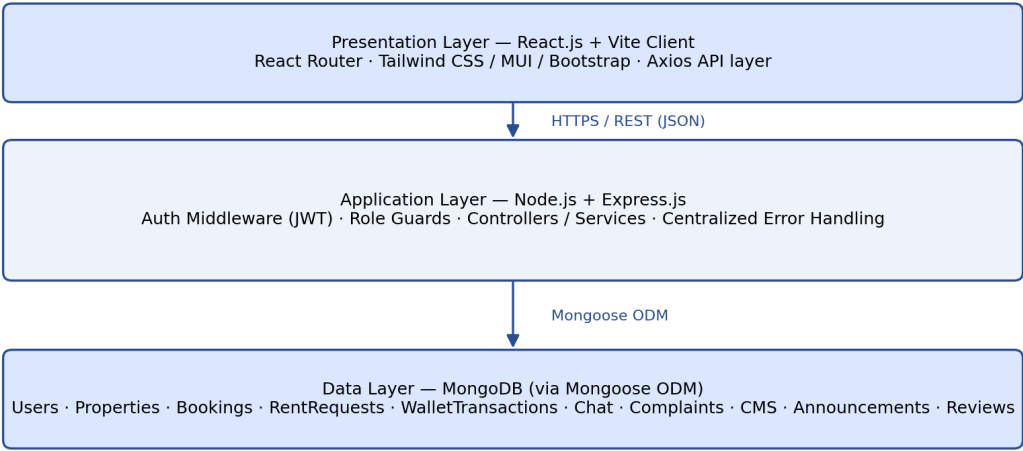


Figure: Three-tier solution architecture — presentation, application and data layers.

5. Project Planning & Scheduling

5.1 Project Planning

Product backlog (representative entries):

Sprint	Epic	User Story	Story Points	Priority	Team Members
Sprint-1	Registration	Register by entering email, password & confirming password	2	High	Vamsi Talari, Chanukya Naidu Abothula
Sprint-1	Login	Log into the application with email & password	1	High	Chanukya Naidu Abothula, Karri Rajeswara Reddy
Sprint-2	Dashboard	View visits requested, confirmed visits and pending answers	1	Medium	Revathi Devi Kota, Vamsi Talari
Sprint-2	Property Management	Add, edit, publish or draft listings with images and amenities	3	High	Karri Rajeswara Reddy, Chanukya Naidu Abothula
Sprint-3	Booking & Negotiation	Request a tour and submit a rent negotiation offer	3	High	Shaik Mohammad Sameer, Karri Rajeswara Reddy

Sprint tracker:

Sprint	Total Story Points	Duration	Start Date	End Date (Planned)
Sprint-1	20	6 Days	—	—
Sprint-2	20	6 Days	—	—
Sprint-3	20	6 Days	—	—
Sprint-4	20	6 Days	—	—

Actual sprint dates and completed story points are pending real execution tracking and are not fabricated here.

6. Functional and Performance Testing

6.1 Performance Testing

Sample functional test scenarios (User Acceptance Testing):

Test Case ID	Scenario	Expected Result
TC-001	Renter registers, logs in, and views a personalized dashboard	Dashboard loads and shows correct visit/booking counts
TC-002	Booking status transitions correctly between Pending, Confirmed and Cancelled	Status updates and reflects on the dashboard
TC-003	Owner listing stays hidden from public search until Admin approval	Listing hidden before approval, visible after
TC-004	Wallet ledger correctly calculates commission and GST	Ledger shows 5% commission and 18% GST on each transaction
TC-005	Complaint submission appears in the Admin complaint centre with correct priority	Complaint appears in queue with correct priority tag

Sample AI-assisted moderation test scenarios (Functional & Performance):

Test Case ID	Scenario	Expected Result
FT-03	Listing moderation scoring for a newly submitted property	Listing receives a spam/validation score and is flagged for Admin review
PT-01	Response time for scoring a submitted listing	Should be under 3 seconds

Actual/Pass-Fail results are intentionally left blank in the working templates until formal UAT execution is carried out.

7. Results

7.1 Output Screenshots

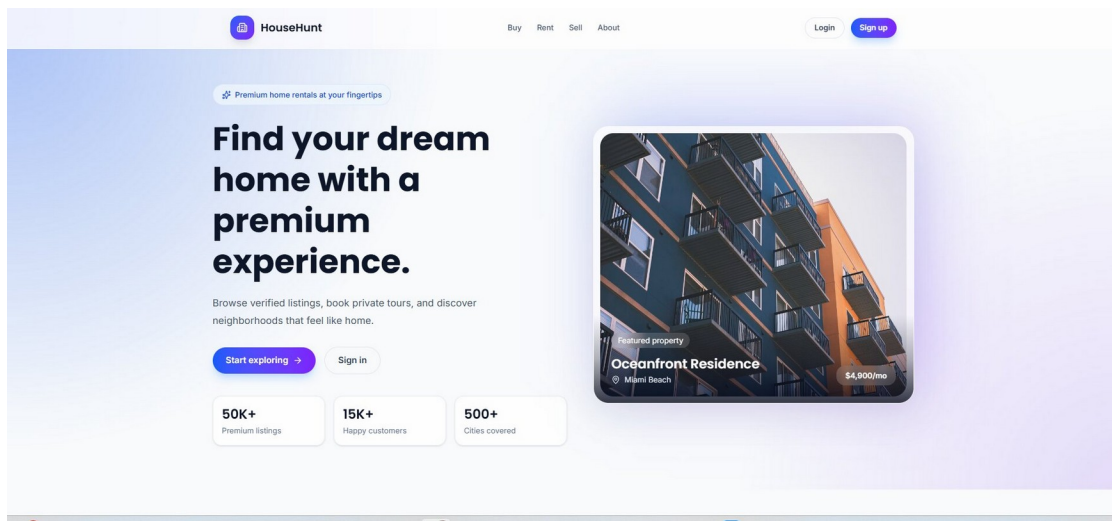


Fig 1 - HouseHunt landing page hero section

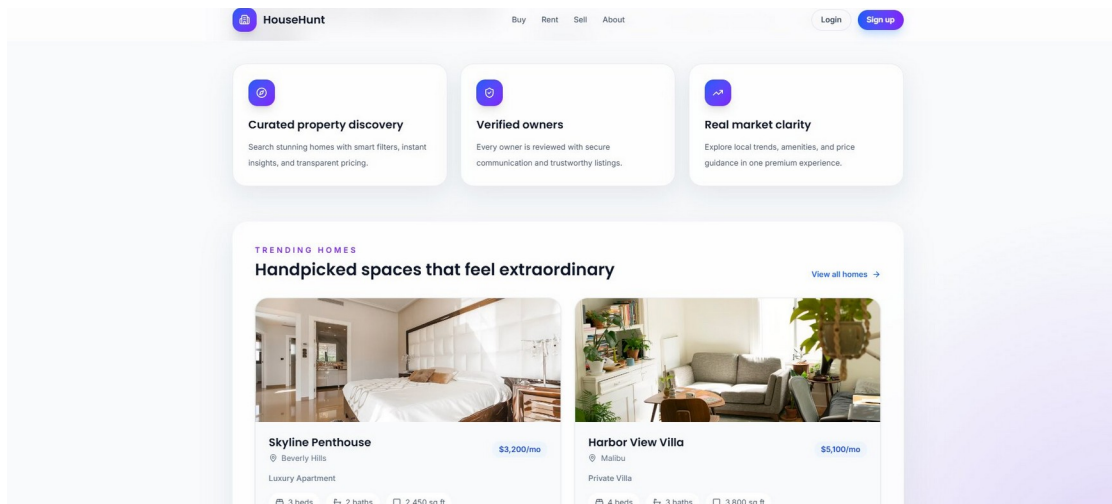


Fig 2 - Platform highlights and trending homes section

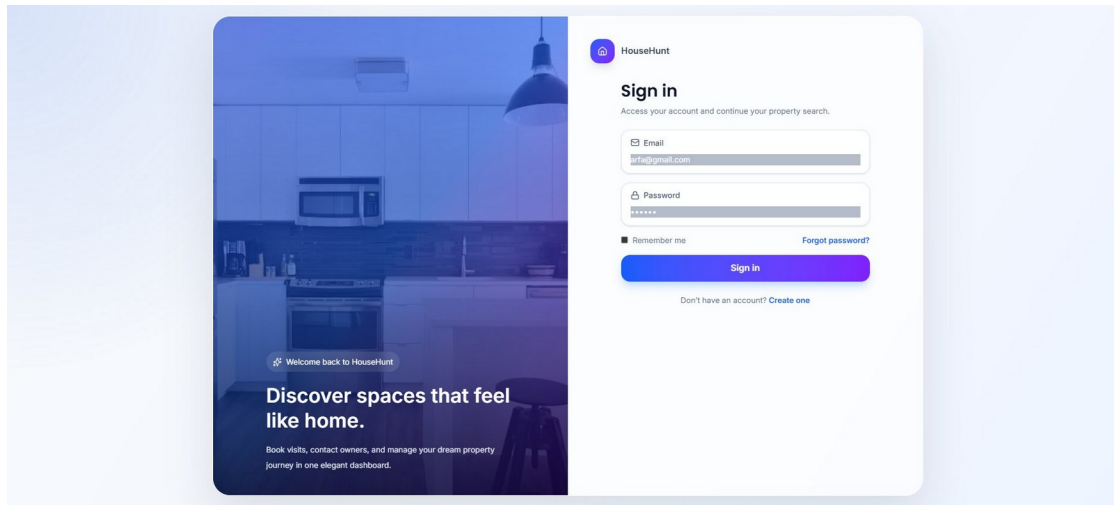


Fig 3 – Renter sign-in page

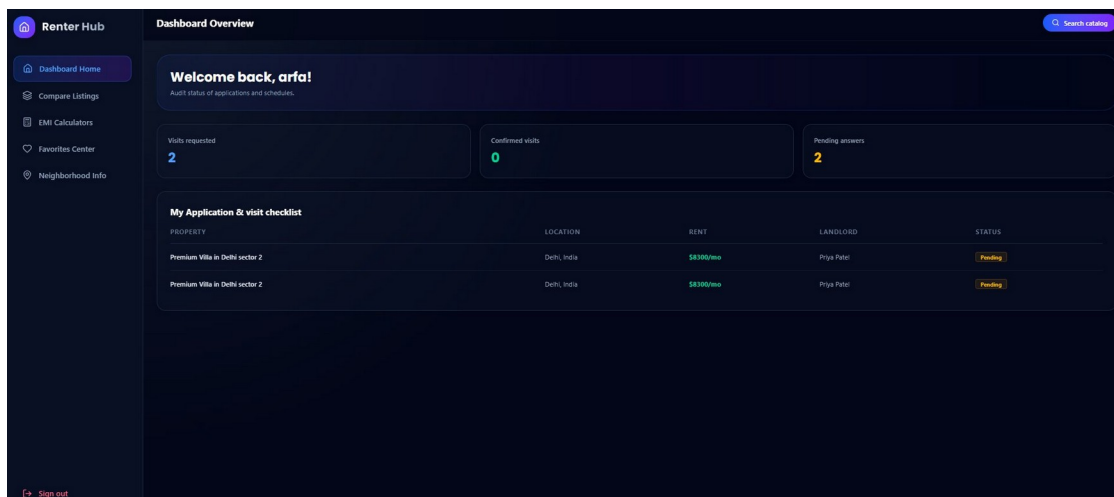


Fig 4 – Renter Hub dashboard overview

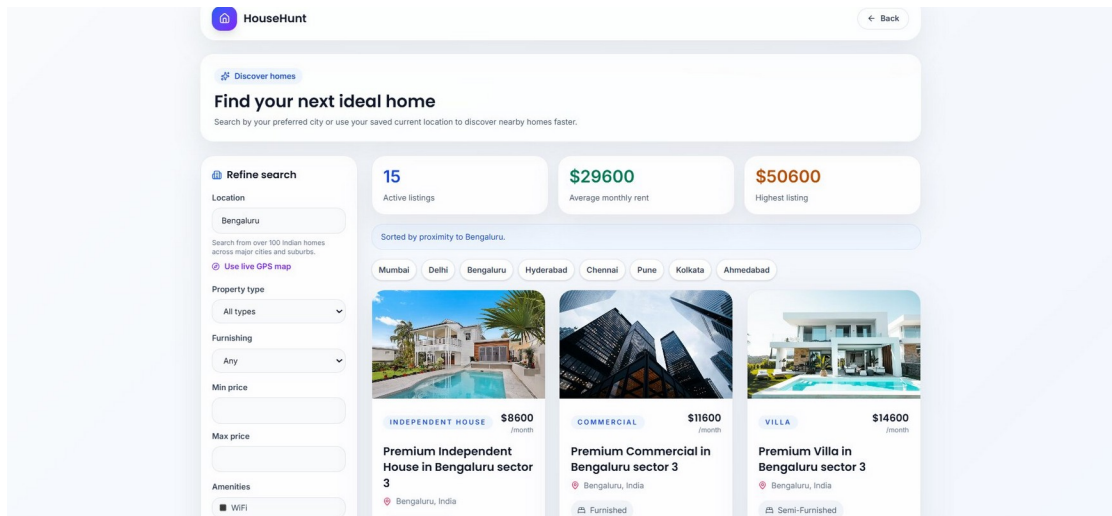


Fig 5 – Property search and filter interface

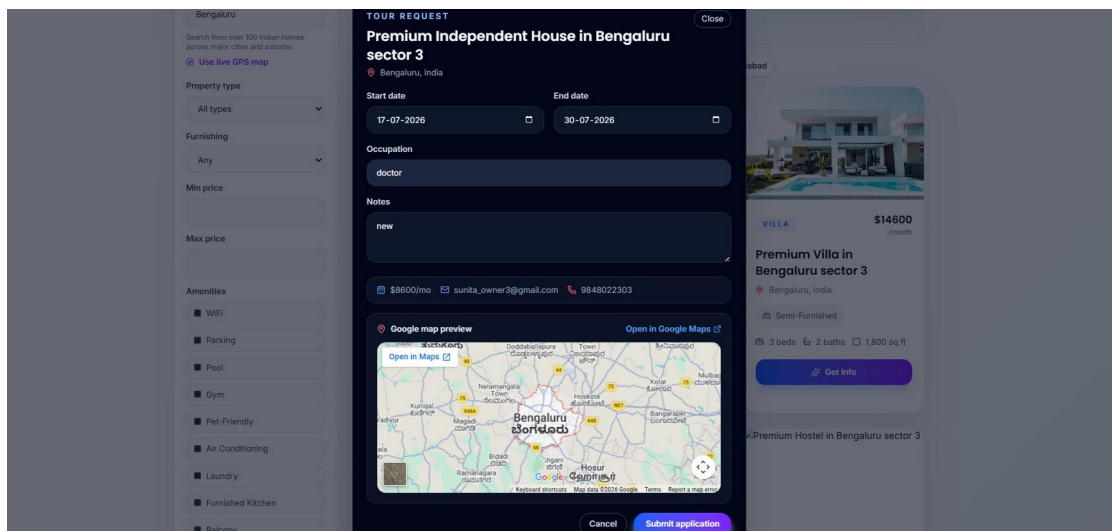


Fig 6 – Tour request modal with Google Maps preview

Renter Hub

Dashboard Home
Compare Listings
EMI Calculators
Favorites Center
Neighborhood Info

Calculators Overview

Mortgage & Rent Yield Calculator

Estimate monthly finance costs and project homeownership returns.

Mortgage Parameters

Property Purchase Price (\$)
300000
Down Payment (\$)
60000
Interest Rate (%)
4.5

Estimated Monthly Mortgage Payment (EMI)

\$1216.04/mo

Based on a 30-year fixed term mortgage with a \$240000 principal loan amount.

Fig 7 – Mortgage & rent yield (EMI) calculator

Renter Hub

Dashboard Home
Compare Listings
EMI Calculators
Favorites Center
Neighborhood Info

Dashboard Overview

Welcome back, arfa!

Audit status of applications and schedules.

Visits requested

4

Confirmed visits

1

Pending answers

0

My Application & visit checklist

PROPERTY	LOCATION	RENT	LANDLORD	STATUS
Premium Villa in Delhi sector 2	Delhi, India	\$8300/mo	Priya Patel	Cancelled
Premium Villa in Delhi sector 2	Delhi, India	\$8300/mo	Priya Patel	Cancelled
Premium Villa in Delhi sector 2	Delhi, India	\$8300/mo	Priya Patel	Cancelled
Premium Villa in Delhi sector 2	Delhi, India	\$8300/mo	Priya Patel	Confirmed

Fig 8 – Renter dashboard showing booking status history

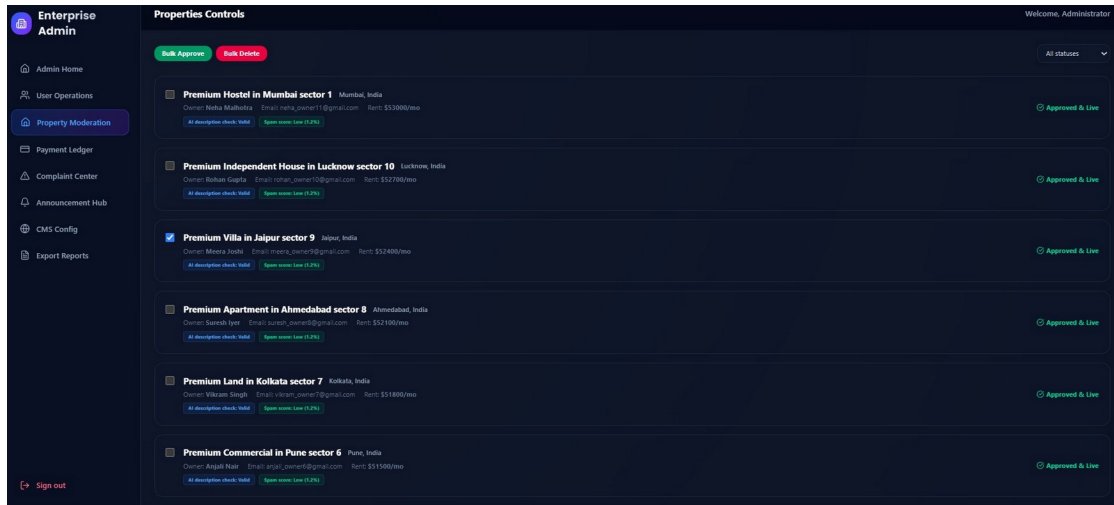


Fig 9 – Admin property moderation console

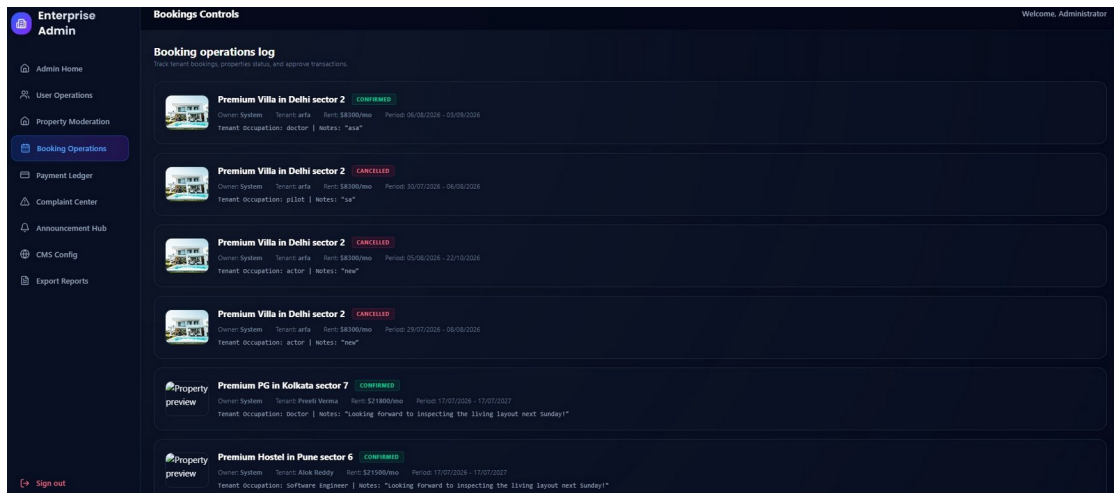


Fig 10 – Admin booking operations log

Enterprise Admin

Admin Home

User Operations

Property Moderation

Booking Operations

Payment Ledger

Complaint Center

Announcement Hub

CMS Config

Export Reports

Sign out

Payments Controls

Platform Transaction ledger

Overview commission collections, GST logs, and payouts logs.

Total Collected Commission (5%)

\$21740.00

Total GST Processed (18%)

\$9009.00

Platform Volume

\$434800.00

Active Transaction Log

Transaction ID	Recipient/Owner	Description	Gross Amount	Collected Commission	GST Tax	Status	Date
TXW-ADM-1794188861481	Anjali Nair	Rent payment processed by Admin for property Premium Hostel in Pune sector 6 (Renter: Ansh Reddy)	\$21500.00	\$1075.00	\$1870.00	Completed	15/07/2028
TXW-ADM-1794187991885	Priya Patel	Rent payment processed by Admin for property Premium Villa in Delhi sector 2 (Renter: arif)	\$8300.00	\$415.00	\$1494.00	Completed	15/07/2028
TXW-2026-4828	John Carter	Rent deposit and advance invoice bn	\$25000.00	\$1250.00	\$225.00	Completed	12/07/2026
TXW-2026-4819	Karan Mehta	Rent deposit and advance invoice bn	\$24500.00	\$1225.00	\$203.50	Completed	12/07/2026
TXW-2026-4818	Divya Bose	Rent deposit and advance invoice bn	\$24000.00	\$1200.00	\$216.00	Completed	12/07/2026
TXW-2026-4816	Shweta Teja	Rent deposit and advance invoice bn	\$23000.00	\$1150.00	\$207.00	Completed	12/07/2026
TXW-2026-4817	Manoj Tiwari	Rent deposit and advance invoice bn	\$23500.00	\$1175.00	\$211.50	Completed	12/07/2026
TXW-2026-4815	Harish Mishra	Rent deposit and advance invoice bn	\$22500.00	\$1125.00	\$202.50	Completed	12/07/2026
TXW-2026-4814	Kavita Choudhary	Rent deposit and advance invoice bn	\$22000.00	\$1100.00	\$198.00	Completed	12/07/2026
TXW-2026-4813	Anil Puri	Rent deposit and advance invoice bn	\$21100.00	\$1055.00	\$195.50	Completed	12/07/2026
TXW-2026-4811	Sanjay Rao	Rent deposit and advance invoice bn	\$20500.00	\$1025.00	\$184.50	Completed	12/07/2026

Fig 11 – Admin payment / commission ledger

Enterprise Admin

Admin Home

User Operations

Property Moderation

Booking Operations

Payment Ledger

Complaint Center

Announcement Hub

CMS Config

Export Reports

Sign out

Complaints Controls

Welcome, Administrator

Complaint & Spam center

Monitor recent flag logs for wrong information, fake listings, or abuse

Type: Wrong Information

Reported By: Vikas Sharma

"The distance to the metro is listed as 500m but it is actually 2.5km."

Status: Open

Resolve Complaint

Medium Priority

Type: Wrong Information

Reported By: Tanish Kumar

"The distance to the metro is listed as 500m but it is actually 2.5km."

Status: Open

Resolve Complaint

Medium Priority

Type: Wrong Information

Reported By: Tanish Kumar

"The distance to the metro is listed as 500m but it is actually 2.5km."

Status: Open

Resolve Complaint

Medium Priority

Type: Wrong Information

Reported By: Swati Mathuraya

"The distance to the metro is listed as 500m but it is actually 2.5km."

Status: Open

Resolve Complaint

Medium Priority

Type: Wrong Information

Reported By: Swati Mathuraya

"The distance to the metro is listed as 500m but it is actually 2.5km."

Status: Open

Resolve Complaint

Medium Priority

Fig 12 – Admin complaint and spam moderation center

8. Advantages & Disadvantages

Advantages

- Single platform for renters, owners and administrators, replacing scattered, manual workflows.
- Admin-approved, verified listings reduce fraud and build renter trust.
- Built-in negotiation, real-time chat and wallet with automatic GST/commission calculation streamline the entire rental transaction.
- Role-based access control (JWT + RBAC) keeps renter, owner and admin capabilities securely separated.
- Modular three-tier architecture and serverless deployment make the platform straightforward to scale and extend.

Disadvantages / Current Limitations

- Online payment gateway is not yet integrated; wallet transactions are currently simulated.
- Real-time chat does not yet support read-receipts or typing indicators.
- Google Maps preview requires an active internet connection and a valid Maps API key in production.
- Bulk property approval does not support partial rollback if one listing in the batch fails validation.
- Push and email notifications for booking status changes are pending implementation.

9. Conclusion

HouseHunt is a complete MERN-stack web application that modernizes the house rental process by enabling secure, efficient interaction between renters, owners and administrators. By combining authentication, role-based dashboards, property management, booking workflows, communication tools, financial tracking and administrative controls into a single scalable platform, HouseHunt lays a strong foundation for evolving into a comprehensive, AI-assisted smart rental ecosystem.

10. Future Scope

- AI-powered property recommendation engine based on renter search behaviour.
- Integrated online payment gateway for rent and deposit collection.
- Native mobile application (iOS & Android).
- Push and email notification system for booking and negotiation updates.
- Video calling for remote property tours.
- AI chatbot for renter and owner support.
- Advanced property analytics dashboard for owners and administrators.

11. Appendix

Source Code

Server: server/ (Node.js + Express, models / controllers / routes / middleware / seed)

Client: client/ (React + Vite, components / pages / context / hooks / services / routes)

Dataset Link

Not applicable — HouseHunt uses live application data (properties, bookings, wallet transactions) seeded automatically into MongoDB on first run, rather than a static training dataset.

GitHub & Project Demo Link

GitHub repository: [github.com/<org>/househunt](https://github.com/your-org/househunt)

Deployed application: refer to the Vercel deployment URL shared alongside this documentation.

Demo accounts — Admin: shaiksameer@gmail.com / admin123 · Owner: john_owner@gmail.com / password123 · Renter: alice_renter@gmail.com / password123